

FindMyNotes: Solving the Student Note Crisis

Students frequently lose track of notes from various subjects and sources, making it difficult to find relevant information quickly for exams or assignments.

We need a centralized platform to **store, organize, and retrieve notes efficiently.**

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PROJECT OVERVIEW

Introducing FindMyNotes

FindMyNotes is a web application designed to allow users to upload, categorize, and search notes easily. Users can search by subject, topic, or keywords to retrieve notes instantly.

- 📄 Our primary goal is to **save time and increase productivity** for students by centralizing their study materials.



Technology Stack: The Foundation

Our application is built on a robust and modern technology stack to ensure dynamic performance, scalability, and security.



Frontend: React.js

Dynamic and responsive user interface for a seamless experience.



Backend: Node.js/Express.js

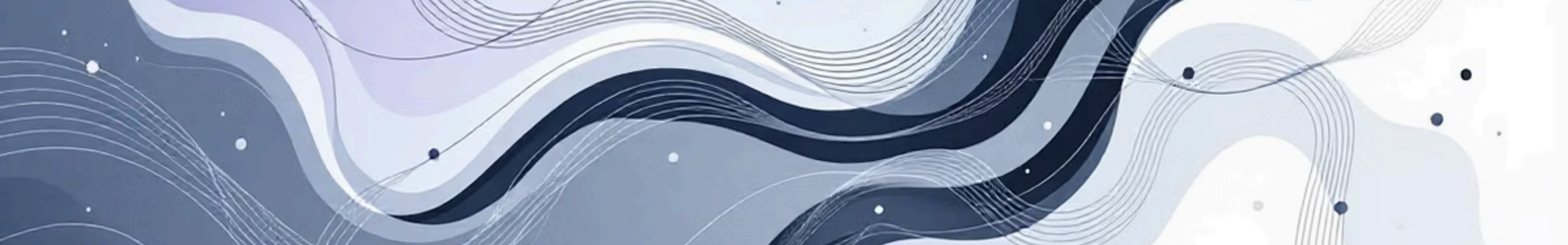
Handles RESTful API requests and business logic efficiently.



Database: MongoDB Atlas

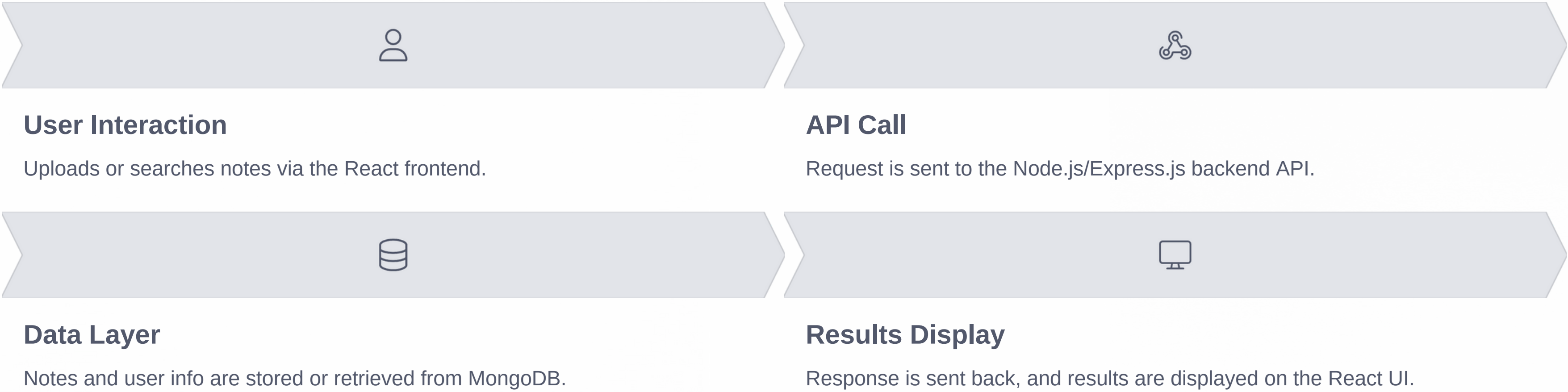
Flexible NoSQL database for storing notes metadata and user information.

Additional tools include **multer** for file uploads, **bcrypt** for secure password hashing, and **cors** for cross-origin handling.



System Architecture & Data Flow

The system follows a clear, three-tier architecture, ensuring reliable data processing from user interaction to database storage.





Core Modules & Key Features

Authentication Module

Secure signup and login functionality with encrypted password storage.

Notes Management

Allows users to upload, categorize (by subject/topic), and delete their notes.

Search Module

Powerful search capability by keyword, topic, or subject for instant retrieval.

Admin Module

Dedicated interface for administrators to manage users and notes across the platform.

RESTful API Endpoints: The Communication Backbone

The backend exposes a set of clear, standardized endpoints for all application functions, ensuring reliable data exchange.

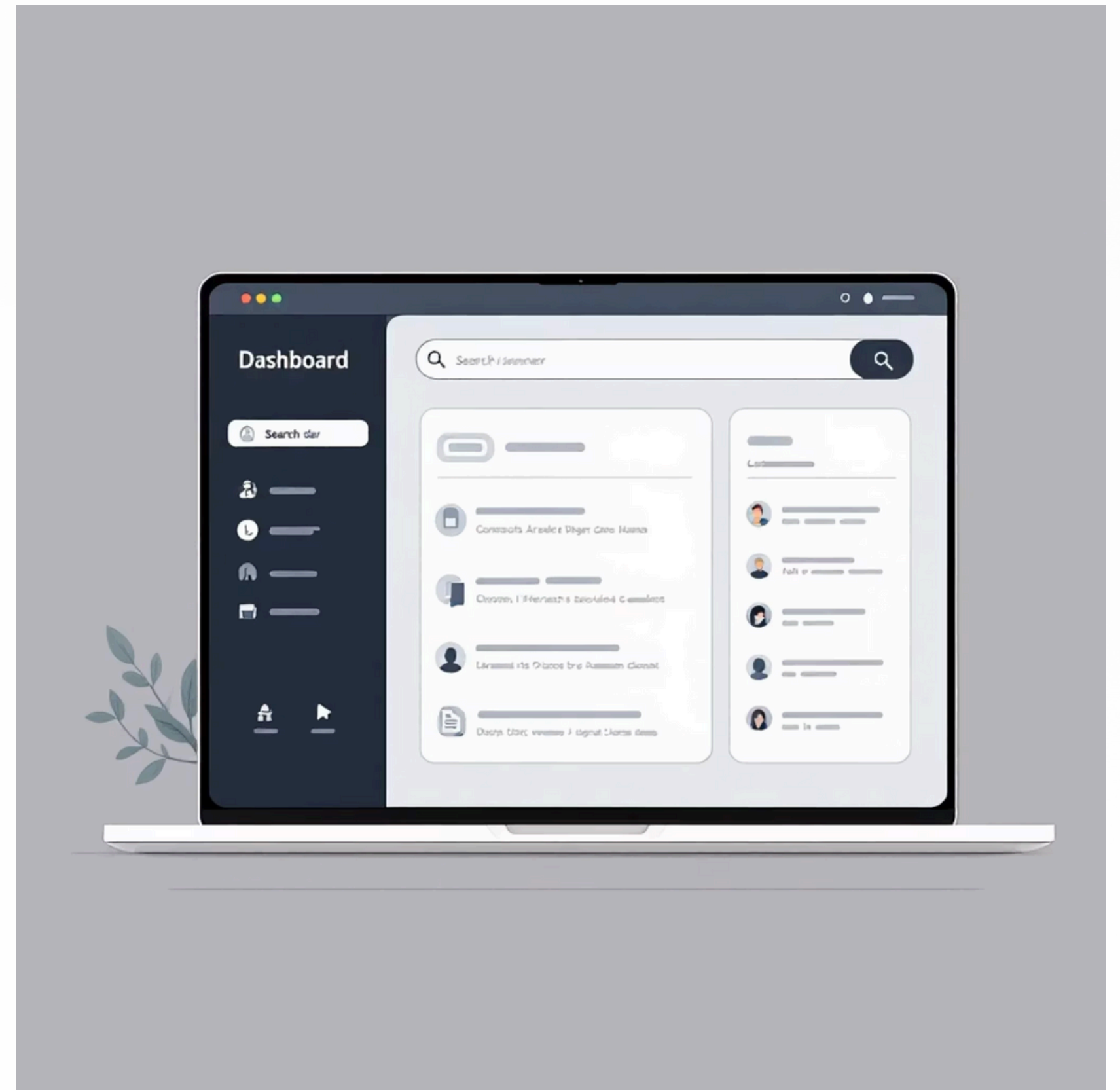
Method	Endpoint	Description
POST	/signup	Register a new user
POST	/login	Authenticate user credentials
POST	/notes/upload	Upload a new note file
GET	/notes/search	Search notes by keyword/filter
DELETE	/notes/:id	Delete a note by its ID

Frontend Integration with React

The user interface is composed of specialized React components, ensuring modularity and maintainability:

- Login and Signup forms
- Dashboard for an overview of user notes
- Upload Notes component for file submission
- Search Notes interface for querying the database

We use **Axios** to call backend APIs, enabling dynamic rendering of search results and user-specific notes in real-time.



Verification & Results

Comprehensive testing confirmed all core functionalities are working accurately and securely.



Authentication

Signup/login tested; password hashing verified for security.



Note Management

Upload, categorization, and retrieval processes successfully verified.



Search Accuracy

Search feature returns relevant results based on keywords and filters.



Admin Control

Admin module successfully manages user accounts and notes.



Conclusion & Future Work

FindMyNotes successfully simplifies note management and retrieval, significantly improving student productivity.



Mobile App Version

Develop native applications for iOS and Android.



Notes Sharing

Implement functionality for users to securely share notes with peers.



AI Keyword Suggestions

Integrate AI for smarter, context-based search and categorization.

References & Resources



The project was developed using official documentation and industry-standard resources, ensuring best practices in development and integration.

- [React.js Documentation](#)
- [Node.js & Express.js Documentation](#)
- [MongoDB Atlas Documentation](#)
- [TutorialsPoint / W3Schools](#) for integration guidance